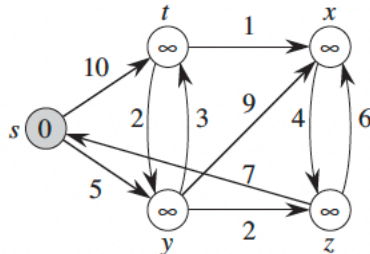
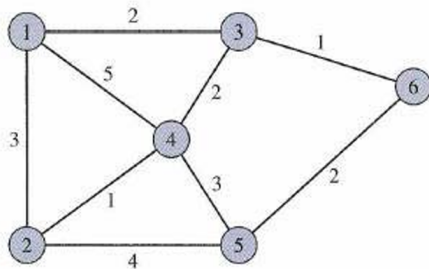


COMP 3270 – Introduction to Algorithms
Homework 10 – Graphs algorithms and Union Data structure, DP

1. Suppose that all edge weights in a graph are integers in the range from 1 to $|V|$. How fast can you make Kruskal's algorithm run? What if the edge weights are integers in the range from 1 to W for some constant W ?
2. Suppose that all edge weights in a graph are integers in the range from 1 to $|V|$. How fast can you make Prim's algorithm run? What if the edge weights are integers in the range from 1 to W for some constant W ?
3. Run Dijkstra's algorithm on the directed graph, first using vertex s as the source and then using vertex z as the source.



4. Give a simple example of a directed graph with negative-weight edges for which Dijkstra's algorithm produces incorrect answers. Show that Dijkstra's does not work when negative-weight edges are allowed?
5. Use Bellman ford algorithm to compute the shortest path from the source vertex, assume 1 is the source.



6. Determine an LCS of $\langle 1, 0, 0, 1, 0, 1, 0, 1 \rangle$ and $\langle 0, 1, 0, 1, 1, 0, 1, 1, 0 \rangle$.
7. Suppose you have an implementation of union that is "by-size" and an implementation of find that does not use path compression. Give the parent map (or array) that results from the following sequence: union(1,2), union(3,4), union(3,5), union(1,7), union(3, 12), union(0,9), union(8,10), union(8,9), union(7,4), union(2,9) where the unions are:
 - a) by size
 - b) by height
 - c) by size, but now with path compression
8. Consider the set of all trees of height h that can be constructed by a sequence of "union-by-height" operations. How many such trees are there?
9. Consider a list of cities $c_1, c_2, \dots, c_n$. Assume we have a relation R such that, for any i, j , $R(c_i, c_j)$ is 1 if cities c_i and c_j are in the same province, and 0 otherwise.
 - a. When the cities are stored in the Disjoint Sets ADT, if you are given two cities c_i and c_j , how do you check if they are in the same province?
 - b. Using a Disjoint Sets ADT, write pseudo-code for an algorithm that puts each city in a set such that c_i and c_j are in the same set if and only if they are in the same province.